

Apo-Antigen Feature Cheat Sheet

Per-residue features from antibody-free MD · what each means, and whether it flags antibody hotspots

At a glance — do hotspots score higher or lower? (surface residues only, binary AUROC)

feature	what it measures	hotspots	AUROC
Flexibility / geometry			
RMSF	how much the residue's Cα moves	↑ higher	0.58
Trajectory SASA	time-averaged exposed area	- chance	0.51
SASA fluctuation	how much its burial 'breathes'	- chance	0.51
Backbone flexibility	φ/ψ backbone wiggling	↓ lower	0.33
Side-chain flexibility	χ ₁ rotamer mobility	↑ weak	0.53
Collective motion			
PC1 participation	moves with the slowest whole-protein mode	↓ lower	0.32
DCCM coupling	how correlated with the rest of the fold	↑ higher	0.62
Intra-protein energy (gRINN)			
intEn van der Waals	LJ packing energy with own fold	↑ higher	0.65
intEn total	Coulomb + LJ with own fold	↑ higher	0.61
intEn electrostatic	short-range Coulomb with own fold	↑ higher	0.61
Hydrogen bonding			
H-bonds to water	H-bonds to solvent (desolvation)	↓ lower	0.34
H-bonds within protein	internal H-bond network	↑ higher	0.57

0.50 = coin flip. ↑ = hotspots higher ↓ = hotspots lower - = no separation.

Better metric — correlation with the graded ΔΔG (not a threshold), per-antigen, leakage-controlled

Using the full binding-energy scale reorders things: hydration leads, interaction energy goes flat.

HB-water + PC1 (combination)	−0.60	higher feature → lower ΔΔG
H-bonds to water	−0.48	higher feature → lower ΔΔG
H-bonds within protein	+0.46	higher feature → higher ΔΔG
PC1 participation	−0.31	higher feature → lower ΔΔG
Trajectory SASA	−0.21	flat
Interaction energy (vdW)	+0.08	flat
Hydrophobicity (Kyte-Doolittle)	0.00	flat

The combination (hydration + PC1, equal weight, unfitted) is the session's flagged result — and it does not overfit.

The picture: hotspots are anchored, pre-organized, DESOLVATED surface patches — less hydrated and decoupled from the slowest collective motion, yet more internally bonded. All effects are weak (N=6 antigens); treat the ranking as hypotheses, not a validated predictor. Page 2–3: the math for each feature.

Reference — how each feature is computed

Notation: r_i = residue i C α position, $\langle \cdot \rangle$ = average over frames, Δr = deviation from mean. Discard first 10 ns; z-score per structure.

Flexibility / geometry

RMSF	$\sqrt{\langle r_i(t) - \langle r_i \rangle ^2 \rangle}$ C α mobility about its mean (nm). Align frames first.
Trajectory SASA	$\langle A_i(t) \rangle$ Shrake–Rupley area, 1.4 Å probe, time-averaged.
SASA fluctuation	$\sqrt{\langle A_i^2 \rangle - \langle A_i \rangle^2}$ std of exposed area over time.
Backbone flexibility	$\text{circstd}(\phi_i) + \text{circstd}(\psi_i)$ circular std of backbone torsions.
Side-chain flexibility	$\text{circstd}(\chi_{1,i})$ χ_1 rotamer mobility (not Gly/Ala).

Collective motion

PC1 participation	$P_i^{(1)} = e_i^{(1)} ^2$ from PCA = eigendecomposition of the C α POSITION covariance; mode 1 = slowest motion. High = swings with it, low = a node.
DCCM coupling	$\frac{1}{n-1} \sum_{j \neq i} D_{ij} , \quad D_{ij} = \frac{\langle \Delta r_i \cdot \Delta r_j \rangle}{\sqrt{\langle \Delta r_i ^2 \rangle \langle \Delta r_j ^2 \rangle}}$ mean abs correlation of i 's motion with all residues.

Intra-protein interaction energy — gRINN (residue↔residue within the apo antigen, no antibody)

Per residue	$E_i^{\text{intra}} = \sum_{j \neq i} (E_{ij}^{\text{elec}} + E_{ij}^{\text{vdW}})$ sum of nonbonded pair energies with own fold (Eq. 38).
elec / vdW	$E^{\text{elec}} = \sum \frac{q_a q_b}{4\pi\epsilon_0 r}, \quad E^{\text{vdW}} = \sum 4\epsilon [(\sigma/r)^{12} - (\sigma/r)^6]$ elec = short-range Coulomb only (PME reciprocal part not per-pair decomposable).

Hydrogen bonding — geometric, counted per frame

Criterion	$D \cdots A \leq 3.5 \text{ \AA} \quad \text{and} \quad D-H \cdots A \geq 150^\circ$ donor D (N,O,S)+polar H, acceptor A. Donor/acceptor atom names are FORCE-FIELD specific (amber H is 'H').
HB to water	$\langle \# \text{ protein} \text{--} \text{water H--bonds} \rangle$ desolvation. NOT hydrophobicity (KD flat) and NOT SASA (r=+0.19).
HB within protein	$\langle \# \text{ protein} \text{--} \text{protein H--bonds} \rangle$ internal network; explicit AMBER selections (guesser undercounts 3–4×).

Two ways to score

Binary AUROC	among surface residues, P(hotspot > rest). Does it flag the top hotspots ($\Delta\Delta G \geq 1$)?
Graded ρ	Spearman vs continuous $\Delta\Delta G$. Does it track the whole gradient? Report per-antigen (leakage-controlled).
Combination	$z(\text{HB water}) + z(\text{PC1})$ equal-weight, sign-aligned, UNFITTED $\rightarrow$ graded $\rho = -0.60$. Adding a 3rd feature degrades it.

Still running

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velocity-mode analysis — the rigorous PC1 (positional PCA re-measures RMSF at $r=+0.52$). Re-check:
hotspots at the true soft-mode nodes?